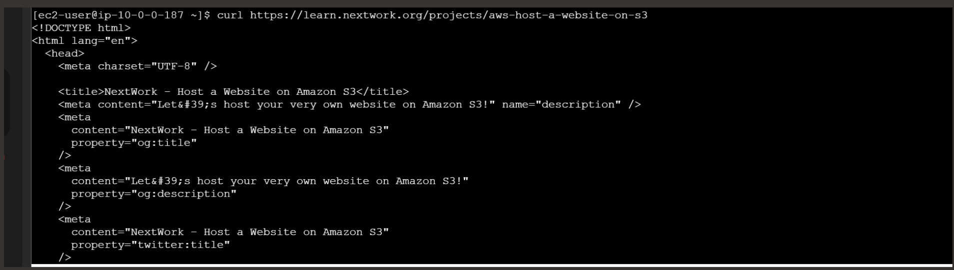
nextwork.org

Testing VPC Connectivity

yakubu abdullahi yusuf



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a service that lets you create your own private network inside AWS where you can run cloud resources like EC2 instances. It is useful because it gives you full control over networking and security.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to configure inbound and outbound rules to ensure reliable connecctivity between my instances and my instaance to the internet.

One thing I didn't expect in this project was...

One thing I didn’t expect in this project was how many different networking and security components such as subnets, route tables, security groups, network ACLs, and gateways have to work together just to enable basic connectivity.

This project took me...

This project took me thirty minutes to complete.

yakubu abdullahi yusuf
NextWork Student

nextwork.org

Connecting to an EC2 Instance

Connectivity is how well different parts of a network talk to each other and with external networks. It's very important because connectivity is also about how data flows smoothly across your network, powering everything from simple web hosting on the Internet to complex operations.

My first connectivity test was whether I could connect to my public instance using AWC Console connect.

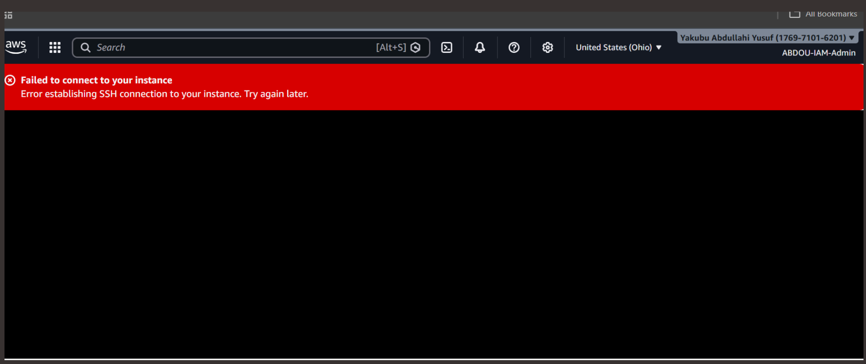
yakubu abdullahi yusuf
NextWork Student

EC2 Instance Connect

I connected to my EC2 instance using EC2 Instance Connect, which is an AWS feature that allows a secure connection to an EC2 instance using SSH directly from the AWS Console or a CLI withput permanently storing SSH keys on the instance.

My first attempt at getting direct access to my public server resulted in an error, because of a rule that is missing in the security group of my public instance.

Are you wondering what rule was missing, I fixed this error by adding an SSH inbound rule with a port 22.

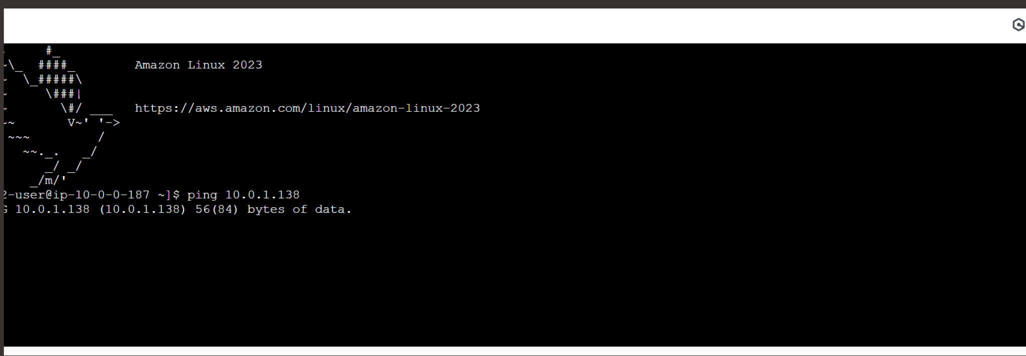


Connectivity Between Servers

Ping is a tool that helps verify connection between two computers or servers. I used ping to test the connectivity between my public EC2 instance and private EC2 instance.

The ping command I ran was ping 10.0.1.138

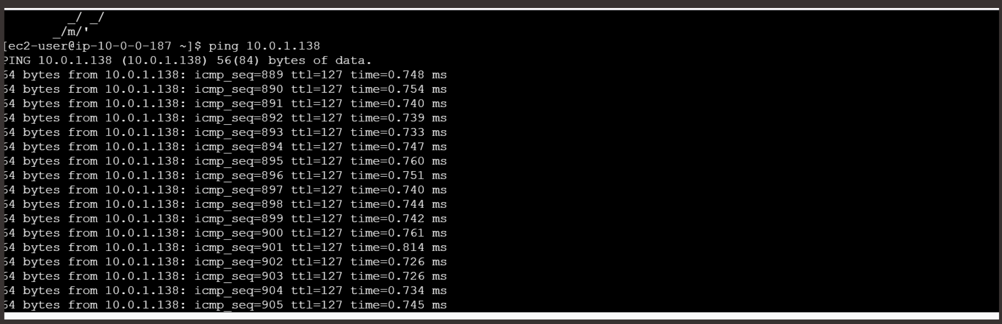
The first ping returned PING 10.0.1.138 (10.0.1.138) 56(84) bytes of data. This meant that the instance could see and reach the private EC2 instance on the network and attempting to send packets to it but couldn't due to an error.



yakubu abdullahi yusuf
NextWork Student

Troubleshooting Connectivity

I troubleshooted this by adding new inbound and outbound rule to my private instance's security group.



Connectivity to the Internet

Curl is used to transfer data to or from a server.

I used curl to test the connectivity between my instance and websites.

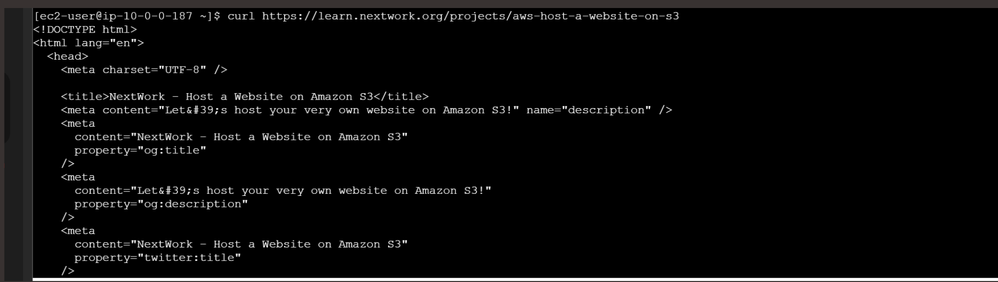
Ping vs Curl

Ping and curl are different because Ping asks if a server is reachable and Curl asks if it can get data from a server.

yakubu abdullahi yusuf
NextWork Student

Connectivity to the Internet

I ran the curl command curl https://learn.nextwork.org/projects/aws-host-a-website-on-s3 which returned the HTML content of the webpage, confirming that my instance had successful internet access and could retrieve data from a web server.



The place to learn & showcase your skills

Check out nextwork.org for more projects